Muhammad Rishal P

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PROFESSIONAL SUMMARY

Computer Science & Engineering undergraduate specializing in systems programming, local AI & RAG architectures, terminal tooling, and embedded IoT. Proficient in Python, TypeScript, C/C++, and Java, with hands-on experience in full-stack architectures, low-level system implementations, and government IT operations.

EDUCATION

Government Engineering College, Kannur **2024 – Present**
B.Tech in Computer Science and Engineering *Kannur, Kerala*

- Coursework: Data Structures, Object-Oriented Programming, Database Systems, Operating Systems.

E.K. Nayanar Memorial Model Polytechnic College **2020 – 2023**
Diploma in Electronics and Communication Engineering (CGPA: 8.77) *Kalyassery, Kerala*

- Strong foundation in digital electronics, networking protocols, and low-level embedded software.

EXPERIENCE

Infosys Springboard **2025 – 2026**
Software Development Intern (Virtual) *Remote*

- Developing a Resume Analysis & Job Recommendation system using Python, Streamlit, and LangChain parsing.
- Implemented secure user authentication with bcrypt and persistent SQLite3 storage.

KELTRON (Kerala State Electronics Development Corporation) **2023**
Network Supervisor & Digital Signature Project Coordinator *Kannur, Kerala*

- Managed IT and network operations at the Department of Treasuries, Government of Kerala.
- Coordinated DSC implementation for secure authentication and government document workflows.

SFO Technologies **2023**
Production Technician – Wire Harnessing *Kochi, Kerala*

- Assembled and inspected high-reliability wire harnesses following industrial quality standards.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, C / C++, Java, SQL, NASM Assembly
- **Systems & AI:** Local LLMs (Ollama GPU), SQLite FTS5 RAG, Piper TTS (ONNX), Web Audio API
- **Backend & Cloud:** FastAPI, Node.js, Express.js, REST APIs, Docker, SQLite (WAL Mode), PostgreSQL
- **Hardware & IoT:** ESP32, Tasmota, MQTT (Mosquitto), Linux (Hyprland / Wayland), Embedded C, Flutter

FEATURED PROJECTS

J.A.R.V.I.S. — Voice AI Companion **github.com/rishalsha/Jarvis**
`Python • Ollama GPU (RTX 3060) • Piper TTS (75ms) • SQLite FTS5 RAG • Web Audio HUD • MQTT`

- Built private voice AI with local GPU inference, 75ms neural TTS, and sub-1ms SQLite FTS5 RAG memory recall.
- Integrated dynamic Web Audio Arc Reactor HUD, real-time barge-in interruption, and MQTT IoT device control.

papr — Terminal Typesetting Engine **Crates.io: papr-cli**
`Rust • CLI • Tectonic • XeTeX • KaTeX`

- CLI tool published on Crates.io compiling Markdown, multi-file LaTeX, and text into publication-grade PDFs.

qmit — AI Conventional Commit CLI **Crates.io: qmit**
`Rust • LLM APIs • Git CLI`

- Open-source CLI published on Crates.io inspecting staged Git diffs to generate standardized conventional commits.

JsonBin — Self-Hosted JSON Engine **github.com/rishalsha/JsonBin**
`FastAPI • Python 3.12 • SQLite WAL • Docker`

- Drop-in replacement for JSONBin.io v3 API with SQLite WAL persistence, Docker packaging, and full CORS control.

Smart Home Automation – Habitat **github.com/rishalsha/Habitat**
`ESP32 • Tasmota • Flutter • MQTT`

- Engineered IoT ecosystem using ESP32 microcontrollers, Tasmota firmware, and Flutter for local IP and voice control.